

Test 4: Volcanoes, Coasts & Running Water

Units 15–17 · pages 212–259 · 50 points · 50 minutes · Closed book

Name _____

Date _____

Score / 50

Read each question carefully. Circle one answer for multiple choice. Write on the lines provided; use the back of the page if you need more room. Point values are shown for each section.

Part A · Multiple Choice

12 questions × 2 points = 24 points

1. Which type of volcano is built almost entirely of loose fragments, giving it very little strength and allowing it to erode quickly?
 - a) Large igneous province
 - b) Shield volcano
 - c) Composite volcano
 - d) Cinder cone
2. The material in a pyroclastic density current can reach temperatures as high as
 - a) 100°C
 - b) 500°C
 - c) 1000°C
 - d) 2000°C
3. What are the two main causes of casualties in volcanic eruptions?
 - a) Earthquakes and tsunamis
 - b) Acid rain and crop failure
 - c) Lava flows and falling ash
 - d) Pyroclastic flows and lahars
4. The 1783–84 Laki eruption in Iceland is estimated to have caused 6 million deaths, chiefly because
 - a) its sulfur dioxide release cooled the Northern Hemisphere, causing crop failures and famine
 - b) its pyroclastic flows swept across Europe
 - c) it triggered a tsunami across the North Atlantic
 - d) its lava flows overran populated valleys
5. As waves approach the shore at an angle, they bend to become more nearly parallel to the beach. This bending is called
 - a) deflation
 - b) reflection
 - c) refraction
 - d) saltation

6. Rip currents
- a) only occur during storms
 - b) carry sediment from the deep ocean onto the beach
 - c) flow parallel to the shore within the surf zone
 - d) flow straight out from shore, returning water that waves have pushed shoreward
7. Which coastline is dominated by *erosional* processes because of relatively recent tectonic activity?
- a) The U.S. Gulf Coast
 - b) The U.S. Atlantic Coast
 - c) The Pacific coast of Canada and the United States
 - d) The Caribbean coast
8. The Bay of Fundy has a daily tidal range as great as 16 m because
- a) it is far deeper than most bays
 - b) its natural oscillation cycle of 12.5 hours matches the frequency of the tides in the adjacent Atlantic
 - c) it lies at a much higher latitude than other bays
 - d) glacial rebound has tilted its floor
9. A *divide* is
- a) a topographically high area that separates one drainage basin from another
 - b) the point where two streams join
 - c) the boundary between fresh and salt water at a river mouth
 - d) a ridge of sediment deposited along a channel
10. The lowest elevation to which a stream can flow is called its
- a) source
 - b) base level
 - c) confluence
 - d) estuary
11. Braided streams develop where
- a) there is more sediment than the stream is able to transport
 - b) the gradient is extremely steep
 - c) the water is unusually clear and free of sediment
 - d) two rivers meet at a confluence
12. Groundwater is often called “fossil water” because
- a) it contains large numbers of microscopic fossils
 - b) it has remained underground for so long, in many cases since the end of the ice ages
 - c) it can only be found in sedimentary rock
 - d) it was formed at the same time as fossil fuels

Part B · Matching

1 question × 8 points = 8 points

13. Write the letter of the correct definition next to each term.

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|----------------------------|--|
| _____ 1. Effusive eruption | a) The area of water over which the wind blows |
| _____ 2. Caldera | b) The volume of water passing a point in a unit of time |
| _____ 3. Fetch | c) A steady, non-violent flow of magma |
| _____ 4. Tombolo | d) Porous rock or sediment that holds and transmits groundwater |
| _____ 5. Berm | e) A volcanic crater more than 2 km in diameter |
| _____ 6. Discharge | f) The eroding outside bank of a stream bend |
| _____ 7. Cut bank | g) A sand deposit connecting the mainland to a near-shore island |
| _____ 8. Aquifer | h) The flatter part of a beach beyond the reach of high tides |

Part C · Short Answer

4 questions × 3 points = 12 points

14. Explain why felsic magma tends to produce explosive eruptions while mafic magma tends to produce effusive ones.

15. Explain how longshore drift moves sediment along a beach.

16. Explain why a meander migrates across its floodplain over time, and how an oxbow lake eventually forms.

17. Name the reservoirs in which Earth's water is stored, and give the approximate proportion held in each.

Part D · Extended Response 1 question × 6 points = 6 points

1 question $\times$ 6 points = 6 points

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